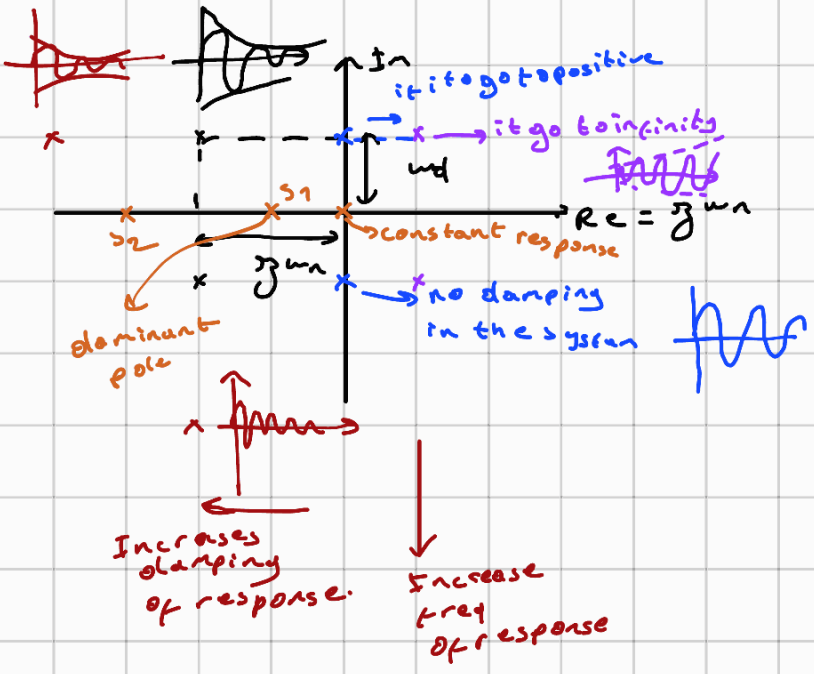
In second order eq

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

- $\zeta < 1$ → $s_{1,2}$ = complex conjugates
- $\zeta = 1$ → $s_{1,2}$ = repeated poles and real
- $\zeta > 1$ → $s_{1,2}$ = distinct real roots



$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$s_{1,2} = -\zeta\omega_n \pm j\omega_d$$

$$e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)$$

Higher order Systems

$$\frac{X(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + b_2 s^{n-2} + \cdots + b_n}{s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_n}$$

- numerator → zeros of the system
- denominator → poles of the system

$m < n$

$$Y(s) = \text{“}\ U(s)$$

Step inputs are mostly used in analysis.

If the poles are real and distinct

$$\frac{X(s)}{U(s)} = \frac{r_1}{s+p_1} + \frac{r_2}{s+p_2} + \cdots + \frac{r_n}{s+p_n} = \sum_{i=1}^{n} \frac{c_i}{s+A}$$

residual

$$r_i = (s+p_i)\left.\frac{X(s)}{U(s)}\right|_{s=p_i}$$

$$X(s) = U(s)\sum_{i=1}^{n}\frac{r_i}{s+p_i}$$

If $U(s)$ is unity (impulse)

$$\mathcal{L}\{x(t)\} = \sum r_i e^{-at}$$

if $U(s) = \frac{1}{s}$

$$Y(s) = \frac{1}{s}\left(\frac{b_0 s^n + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}\right)$$

$$= \frac{r_0}{s} + \frac{r_1}{s+p_1} + \frac{r_2}{s+p_2} + \cdots + \frac{r_i}{s+p_i}$$

$$\hookrightarrow x(t) = a_0 + \sum_{i=1}^{n} r_i e^{-p_i t}$$

if poles are not all real values

for $U(s) = \frac{1}{s}$

$$Y(s) = \frac{a_0}{s} + \sum_{j=1}^{q} \frac{a_j}{s+p_j} + \sum_{k=1}^{2r} \left(\frac{b_k (s+\zeta_k \omega_k) + c_k \omega_{dk}}{s^2 + 2\zeta_k \omega_k s + \omega_k^2}\right)$$

'q' poles are real valued

'2r' poles are complex valued

$q + 2r = n$

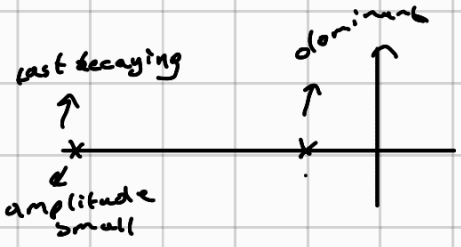
$$x(t) = a_0 + \sum_{j=1}^{q} a_j e^{-p_j t} + \sum_{k=1}^{r} b_k e^{-\zeta_k \omega_k t} \cos \omega_{dk} t + \sum_{k=1}^{r} c_k e^{-\zeta_k \omega_k t} \sin \omega_{dk} t$$

$$\omega_{dk} = \omega_k \sqrt{1 - \zeta_k^2}$$

if $z_n = p_1 \Rightarrow r = 0$

zero and pole cancelation

if poles go must further to the Im axis the residues decrease



whatwe need

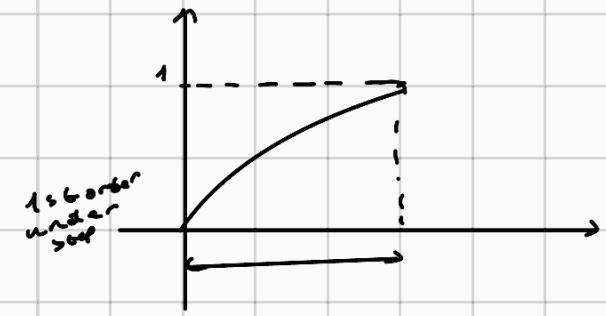
Stability

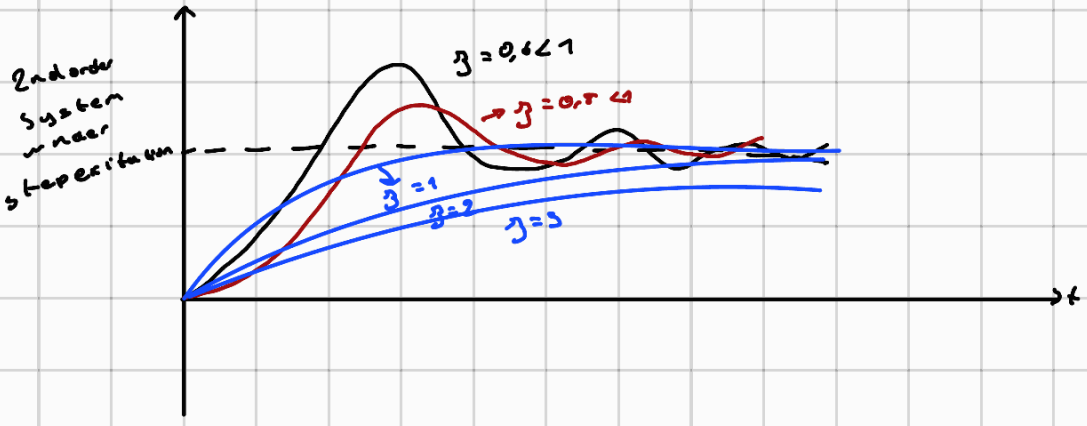
Accuracy (error)

Quality of response

Quality of response → it focus on how we get to the steady state response

* later we'll be covering accuracy and stability of reponse as well.



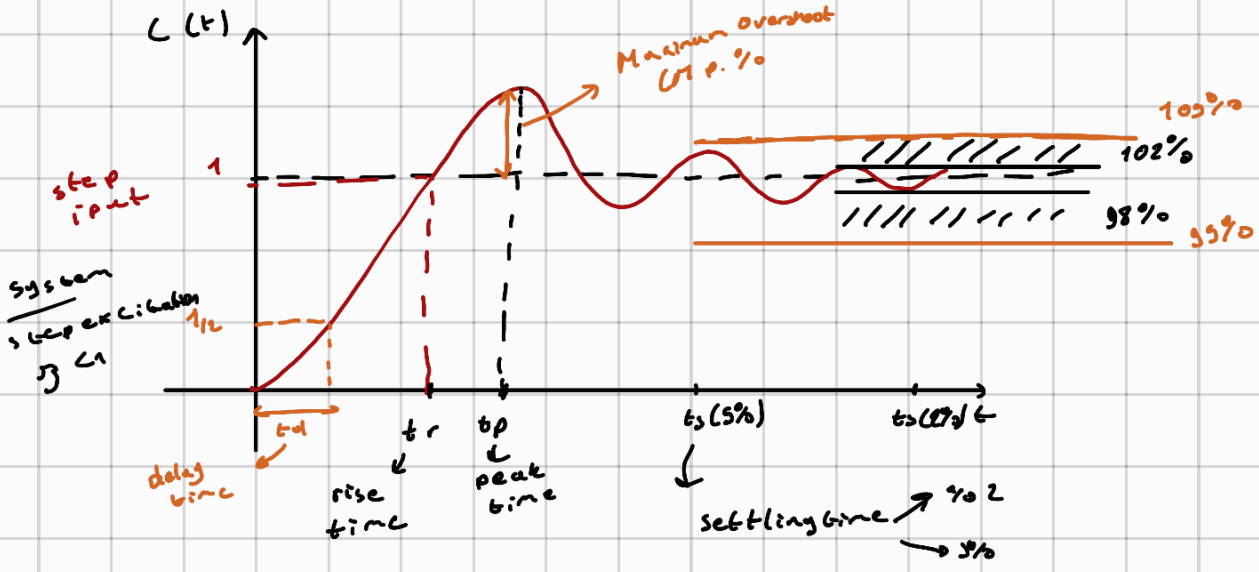


damping : $Re: \zeta\omega_n = \sigma$ sigma

Freq : $Im: \omega_n\sqrt{1-\zeta^2} = \omega_d$

$\zeta = 1$ critically dumped case

settles down (gets to the final response) in the minimum amount of time



$\zeta < 1$: time needed to set from 0% to 100%

$\zeta > 1$: " " " " " 10% to 90%

rise time (t_r)

$$c(t) = 1 - e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)$$

$$1 = \underbrace{1 - e^{-\zeta\omega_n t_r}\left(\cos\omega_d t_r + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t_r\right)}_{0} \Rightarrow \cos\omega_d t_r + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t_r = 0$$

$$1 + \frac{\zeta}{\sqrt{1-\zeta^2}}\tan\omega_d t_r = 0$$

$e^{-\zeta\omega_n t_r}$ can not be zero in finite time

only when $t_r \to \infty \Rightarrow e^{-\zeta\omega_n t_r} \to 0$

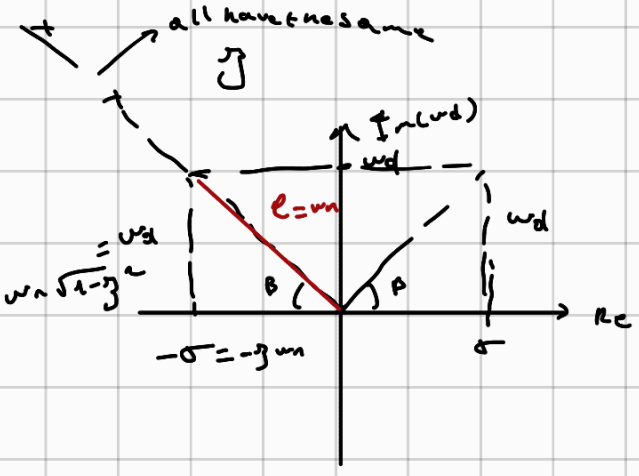
$$\tan\omega_d t_r = -\frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\Rightarrow \omega_d t_r = \tan^{-1}\left(-\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

$$\Rightarrow t_r = \frac{1}{\omega_d}\tan^{-1}\left(\frac{-\omega_n\sqrt{1-\zeta^2}}{\omega_n\zeta}\right)$$

$$= \frac{1}{\omega_d}\tan^{-1}\left(\frac{\omega_d}{-\sigma}\right)$$

$$= \frac{\pi - \beta}{\omega_d} = t_r$$



$$l^2 = (\zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)$$

$$l^2 = \omega_n^2$$

$$\cos\beta = \frac{\zeta\omega_n}{\omega_n} = \zeta$$

$$\beta = \cos^{-1}\zeta$$

peak time (t_p):

$$\frac{dc}{dt} = 0$$

$$\zeta\omega_n e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t_p\right) - e^{-\zeta\omega_n t_p}\left(-\omega_d\sin\omega_d t_p + \frac{\zeta\omega_d}{\sqrt{1-\zeta^2}}\cos\omega_d t_p\right) = 0$$

$$\Rightarrow e^{-\zeta\omega_n t_p}\left(\zeta\omega_n \cos\omega_d t_p + \frac{\zeta^2\omega_n^2}{\omega_n\sqrt{1-\zeta^2}}\sin\omega_d t_p - \omega_d \sin\omega_d t_p - \frac{\zeta\omega_d}{\sqrt{1-\zeta^2}}\cos\omega_d t_p\right)=0$$

$$\left(\frac{\zeta^2\omega_n^2}{\omega_d} + \omega_d\right)\sin\omega_d t_p = 0$$

$$\hookrightarrow \omega_d t_p = \pi, 2\pi, 3\pi, \dots$$

$$\boxed{t_p = \frac{\pi}{\omega_d}}$$

$$c(t_p) = 1 - e^{-\zeta\omega_n \frac{\pi}{\omega_d}}\left(\cos\pi + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\pi\right)$$

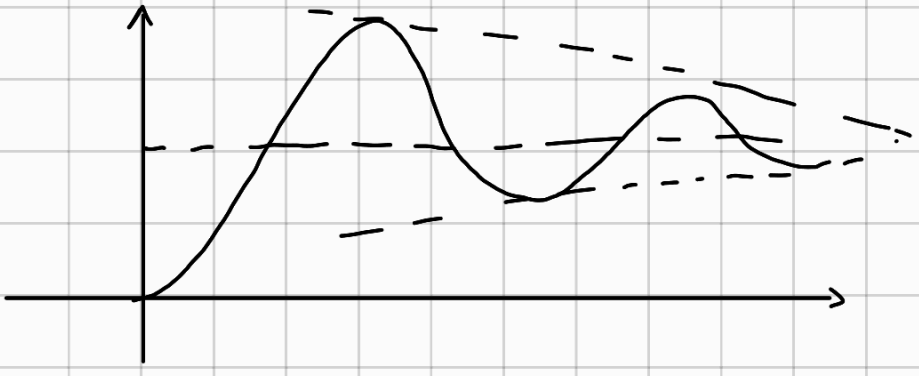
$$= 1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\%M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\% \quad //$$

$$M_p = 1 + e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} - 1 = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = e^{-\frac{\pi\zeta\omega_n}{\omega_n\sqrt{1-\zeta^2}}} = e^{-\frac{1}{\omega_d}\sigma}$$

Settling time (t_s)

$$c(t) = 1 - e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)$$



$$c(t) = 1 - a_1 \sin(\omega_d t + \alpha_2)$$

$$-1 \le \sin \le 1$$

$$= 1 \pm a_1$$

$$\sin(\omega_d + \alpha) = \sin\omega_d + \cos\alpha + \cos\omega_d + \sin\alpha$$

$$c(t) = 1 - e^{-\zeta \omega_n t}\left(\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t\right)$$

$$c(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}}\left(\sqrt{1-\zeta^2}\cos \omega_d t + \zeta \sin \omega_d t\right)$$

$$\sin(\omega_d t + \alpha) = \sin \omega_d t \cdot \cos\alpha + \cos \omega_d t \cdot \sin\alpha$$

$$\begin{cases} \cos\alpha = \zeta \\ \sin\alpha = \sqrt{1-\zeta^2} \end{cases} \rightarrow \tan\alpha = \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\alpha = \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

$$= 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_d t + \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right)$$

bounding curves: $1 \pm \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} = \begin{cases} 1 \mp 0{,}05 & 5\% \\ 1 - 0{,}02 & 2\% \end{cases}$

$$5\% \rightarrow \frac{e^{-\zeta \omega_n t_s}}{\sqrt{1-\zeta^2}} = 0{,}05 \Rightarrow e^{-\zeta \omega_n t_s} = 0{,}05\sqrt{1-\zeta^2}$$

$$\Rightarrow -\zeta \omega_n t_s = \ln 0{,}05 + \ln\sqrt{1-\zeta^2}$$

$$\Rightarrow t_s = -\frac{\ln 0{,}05}{\zeta \omega_n} - \frac{\ln\sqrt{1-\zeta^2}}{\zeta \omega_n} \quad \text{(} \ln\sqrt{1-\zeta^2} \rightarrow 0 \text{)}$$

for small ζ values

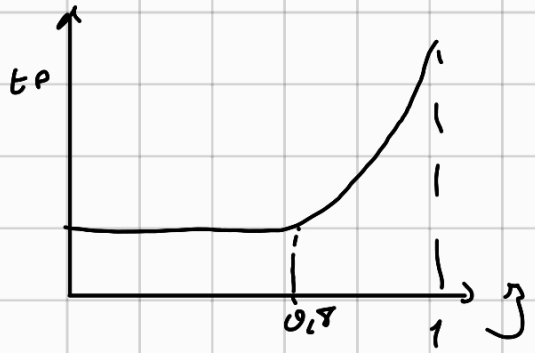
$$\rightarrow t_s = \frac{3}{\zeta \omega_n} = 3\tau$$

because $\frac{1}{\zeta \omega_n}$

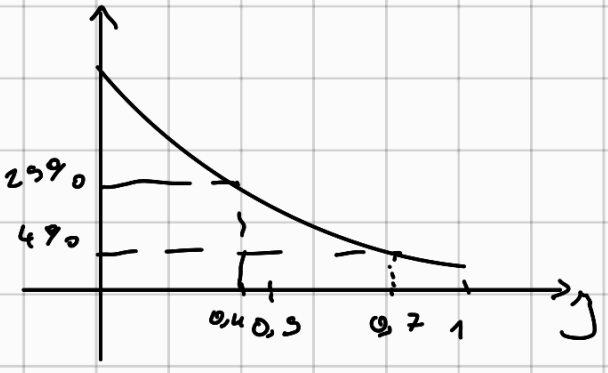
$$t_r = \frac{1}{\omega_d} \tan^{-1}\left(\frac{-\omega_d}{\sigma}\right)$$



$$t_p = \frac{\pi}{\omega_d}$$

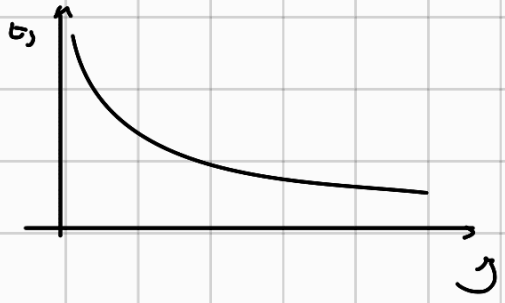


$$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$$



smaller

$$t_s = \frac{3}{\zeta\omega_n} \text{ or } \frac{4}{\zeta\omega_n}$$

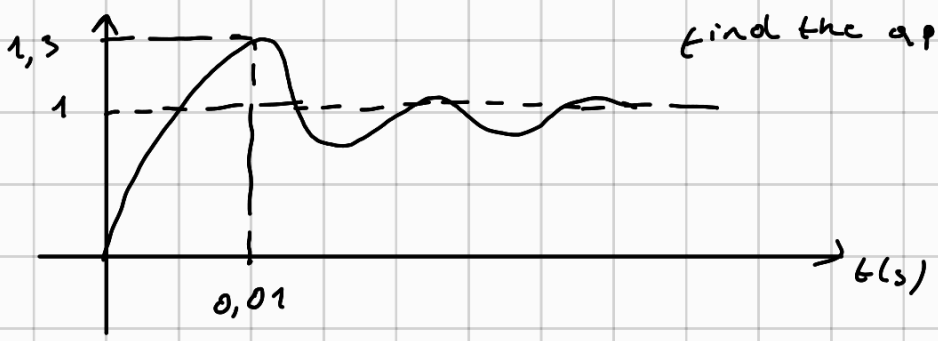


M_p and t_r are contradictory !!

t_d = ? delay time

$t_d = c(\zeta)$

Example: Step response of a 2nd order system is given



find the approximate T.F

$t_p = 0,01$

$M_p = 0,3 = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \Rightarrow \zeta = 0,356$

$t_p = 0,01 = \frac{\pi}{\omega_d} \rightarrow \omega_d = 100\pi \Rightarrow \omega_n\sqrt{1-\zeta^2} = 100\pi$

$\hookrightarrow \omega_n = 336$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} = \frac{(336)^2}{s^2+2(0,356)(336)s+(336)^2}$$

Example $\frac{C(s)}{R(s)} = \frac{k}{s^2+as+k}$ for step input

$k, a = ?$

Maximum overshoot $\leq 10\%$

Setting time ≤ 1 sec (5%)

$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \leq 0,1 \Rightarrow -\frac{\pi\zeta}{\sqrt{1-\zeta^2}} \leq \ln(0,1)$

$\Rightarrow \frac{\pi\zeta}{\sqrt{1-\zeta^2}} \geq \ln(0,1) \Rightarrow \zeta > 0,6$

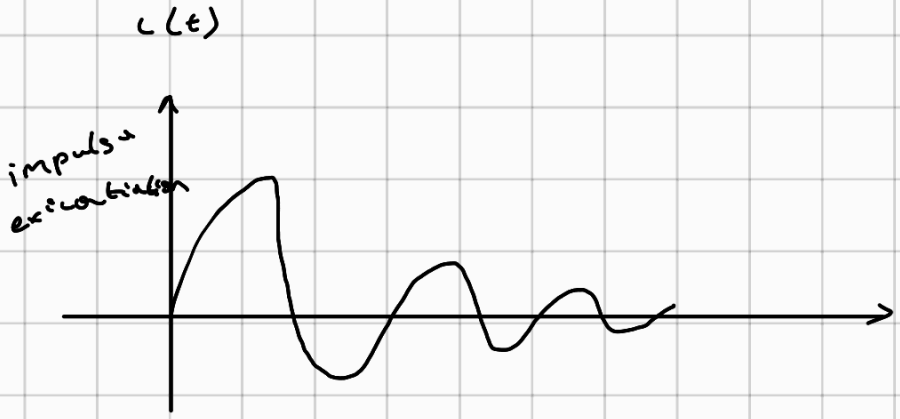
$t_s = \frac{3}{\zeta\omega_n} \leq 1$

$\zeta\omega_n \geq 3$

$\begin{cases} \zeta = 0,7 \\ \zeta\omega_n = 4 \rightarrow \omega_n = \frac{4}{0,7} \end{cases}$

$\begin{cases} \omega_n^2 = k \\ 2\zeta\omega_n = a \end{cases}$

SISO Tool MatLab



Ex:

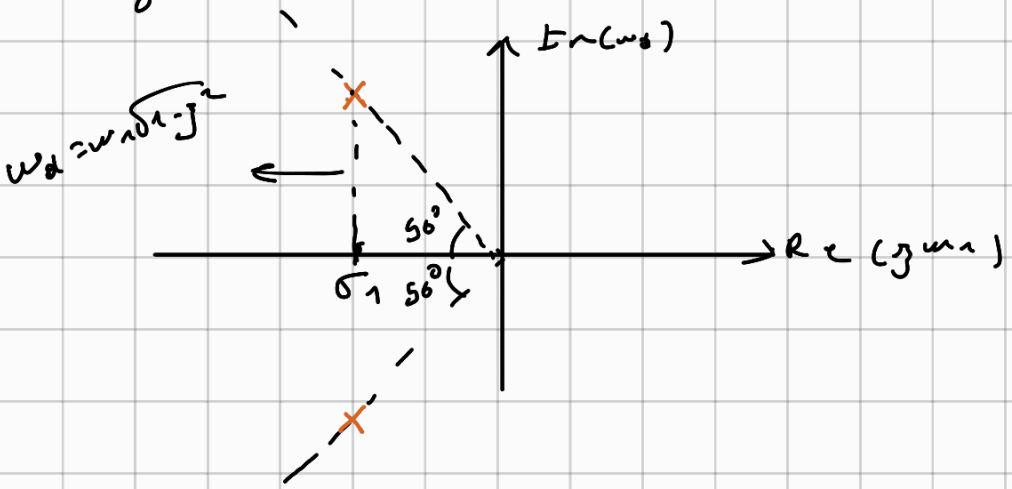
Find the specific location of poles on the splane

a) $M_p = 12\%$, $t_s = 0.6$ sec → stable

b) $M_p = 10\%$, $t_s = 3$ sec

$$M_p = 0.12 = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \Rightarrow \zeta = 0.599$$

$$\%0\ t_s = \frac{3}{\zeta\omega_n} = 0.6 \Rightarrow \omega_n = 8.94 \text{ rad/s}$$



$$\zeta\omega_n = 0.599 \cdot 8.94 = \sigma_1$$

$$\beta = \cos^{-1}\zeta = \cos^{-1} 0.599 = 56^\circ$$

$$p_1 = -\sigma_1 + j\omega_n\sin\beta \quad (\omega_n\sqrt{1-\zeta^2})$$

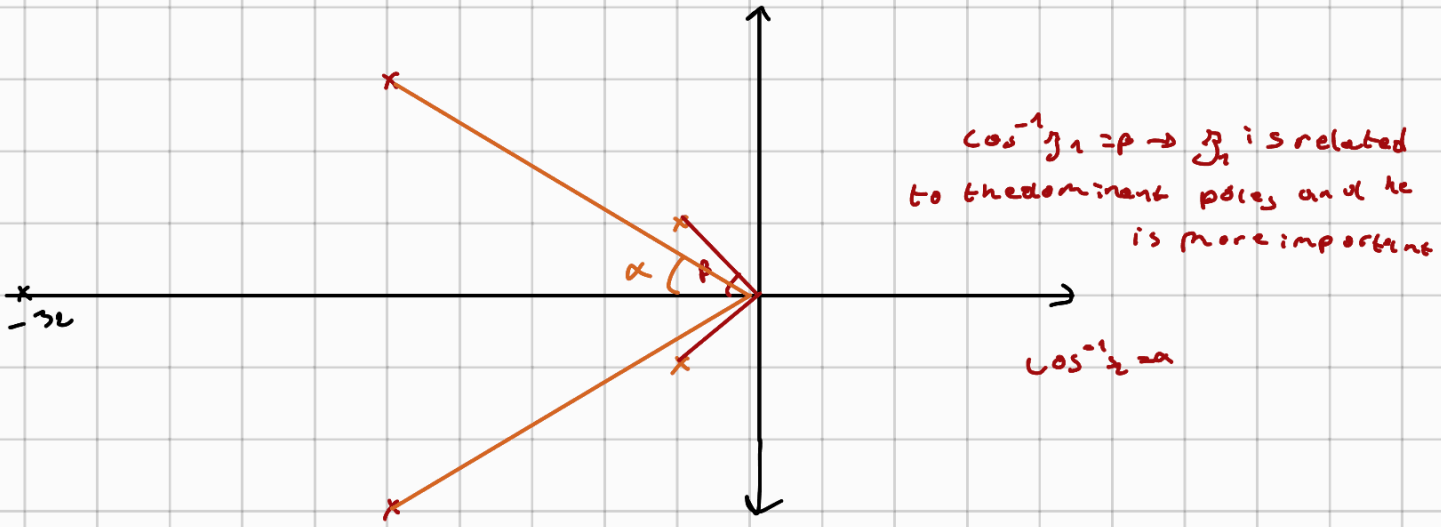
$$p_2 = -\sigma_1 - j\omega_n\sin\beta$$

Find ζ for the given C.L.T.F.

$$G(s) = \frac{1638\,(s^2+2,6s+1,69)}{(s+3,24)\,(s+0,855+j0,672)\,(s+0,855-j0,672)\,(s+5,24+j6,56)\,(s+5,24-j6,56)}$$

(3,24: real pole; 0,855 ± j0,672: c.c. pole; 5,24 ± j6,56: c.c. with negative real part)

with negative real part



$\cos^{-1}\zeta_1 = \beta \rightarrow \zeta_1$ is related to the dominant poles and he is more important

The system is combine of 2 second degree systems

by looking to the dominant poles 2 can be find

$$\tan\beta = \frac{\omega_d}{\sigma} = \frac{0,672}{0,855} \rightarrow \beta = 38,16^\circ \rightarrow \zeta_1 = 0,78$$

$$\tan\alpha = \frac{6,56}{5,24} \rightarrow \alpha = 51,38^\circ \rightarrow \zeta_2 = 0,62$$